



4 The random variable X has probability density function f given by

$$f(x) = \begin{cases} \frac{1}{21}(x-1)^2 & 2 \leq x \leq 5, \\ 0 & \text{otherwise.} \end{cases}$$

(a) Find the cumulative distribution function of X . [3]

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The random variable Y is defined by $Y = (X-1)^4$.

(b) Find the probability density function of Y . [3]

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(c) Find the median value of Y . [2]

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(d) Find $E(Y)$. [2]

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